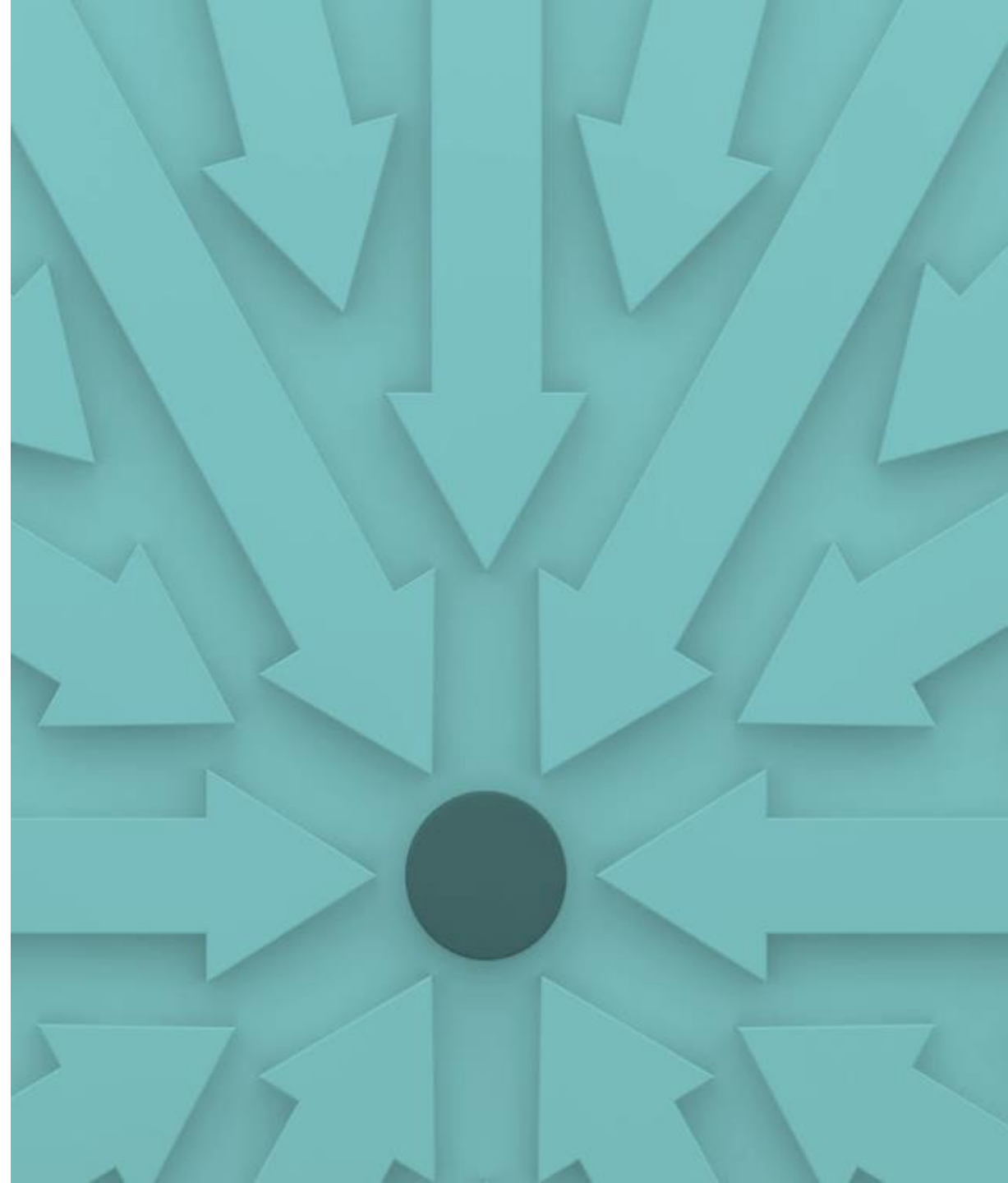




AI-Assisted App Prototyping with Bloom

Dr. Kornelia Papp

The paradigm shift:
moving from
syntax-first to
intent-first.



Agenda

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Session goals

In this session, we explore how vibe coding turns high-level creative intent into working mobile software.

What is Vibe Coding?

Understand how natural language, creative intent, and AI-assisted development can be used to design and build functional native mobile applications.

How does the developer role change?

Analyze the shift from programmer to architect, where engineers increasingly define systems, constraints, interfaces, and product intent rather than writing every line of code.

How can Bloom support app creation?

Use the Bloom development environment to create and customize real-time mobile interfaces without manually writing syntax.

How do we share apps instantly?

Learn how instant deployment technologies, such as App Clips make it possible to distribute native app experiences.

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Vibe Coding

Vibe coding refers to a conversational mode of software development in which the user does not interact with the code directly, but steers an AI coding assistant through natural-language instructions, feedback, and testing.

The development process becomes iterative: the user specifies intent, the AI implements it in code, the user evaluates the result, and further prompts are used to correct, extend, or refine the application.



Andrej Karpathy ✓

@karpathy



There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

12:17 AM · Feb 3, 2025 · **4.1M** Views

Development Flows

- **Traditional coding:** you write the code.
- **Vibe coding:** you specify intent, let AI do most of the implementation, and judge mainly from behaviour and output.
- **AI-assisted coding:** AI suggests code, but you remain fully in control.

Vibe Coding Stack

1. Client Layer (The Interface)

Device

Native iOS / Android app

Role

Captures "user intent" (voice / text / gestures)

Tech

React Native shell that interprets the "vibe"

Intent



2. Orchestration Layer (The Brain)

AI Agent

The orchestrator; plans the app structure

Code Engine

Generates React Native code on the fly

Logic Mapping

Connects UI components to backend actions

Plan + Code



3. Infrastructure Layer (The Cloud)

Cloud Database

Serverless dynamic database; schema evolves with features

API

Hidden GraphQL / REST layer handling data flow

Deployment

App Clips + OTA updates push app to phone

Data + Deploy

Bloom

What is Bloom?

<https://bloom.diy/>

Bloom is for rapid prototyping of small, useful mobile apps.

It is strongest when you want to test an idea fast, validate a workflow, or build an MVP without getting stuck in frontend, backend, and deployment details. That framing is also consistent with Bloom's own messaging around going from idea to production-ready apps quickly.



Startupticker.ch

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Founded this year, [Bloom \(YC X25\)](#) has already drawn tens of thousands of users to its no-code platform, which builds both frontend and backend in under two minutes. It delivers production-ready mobile apps developers can share.

The Swiss-US startup founded by [David Oort Alonso](#) and [Sirian Maathuis](#) has recently secured \$3.4M in pre-seed funding led by [Moonfire](#) with participation from Y Combinator and Pioneer Fund to fast-track growth through hiring.

Learn more about the story behind the headlines in our article: <https://lnkd.in/emTNGAKW>



Bloom Use Case

Start with the **problem**. A good Bloom use case is usually: **small, concrete, repetitive, and valuable**.

- Start with **one specific task**
- Specify the **target user**
- Define the **format of the answer**
- Add **constraints** like length, tone, and what not to do
- Test with both a **normal input** and a **messy input**
- Revise the prompt after each test

Bloom offers a free tier with 350 credits per month, enough for [basic experimentation](#) and small prototypes.

Clear and Simple Workflow

Clear input → transformation → output

Bloom-style apps are easiest when the workflow is simple:

- user enters or uploads something
- app processes it
- app returns something useful

Examples:

- input: messy notes → output: structured summary
- input: task list → output: prioritized plan
- input: symptom log → output: clean timeline
- input: study topic → output: quiz questions

Mobile-first Usefulness

Bloom is positioned around native mobile apps.

- Does this make sense on a phone?
- Would someone use it on the go?
- Does quick access matter?

Strong mobile use cases:

- checklists
- trackers
- quick capture forms
- small decision tools
- campus helper apps

Less ideal for this format:

- giant dashboards
- complex spreadsheets
- long-form editing systems

Use Case Assessment

The 5-question Bloom filter

A use case is good if they can answer “yes” to most of these:

1. Can I explain the app in one sentence?
2. Does it solve one specific problem?
3. Would someone use it on a phone?
4. Can I demo value in under 30 seconds?
5. Can I build a first useful version in this session?

Examples that pass:

- “An app that turns a lecture topic into three oral exam questions.”
- “An app that helps lab students log sample preparation steps.”
- “An app that converts raw meeting notes into action items.”

‘Berufe entdecken’ Application

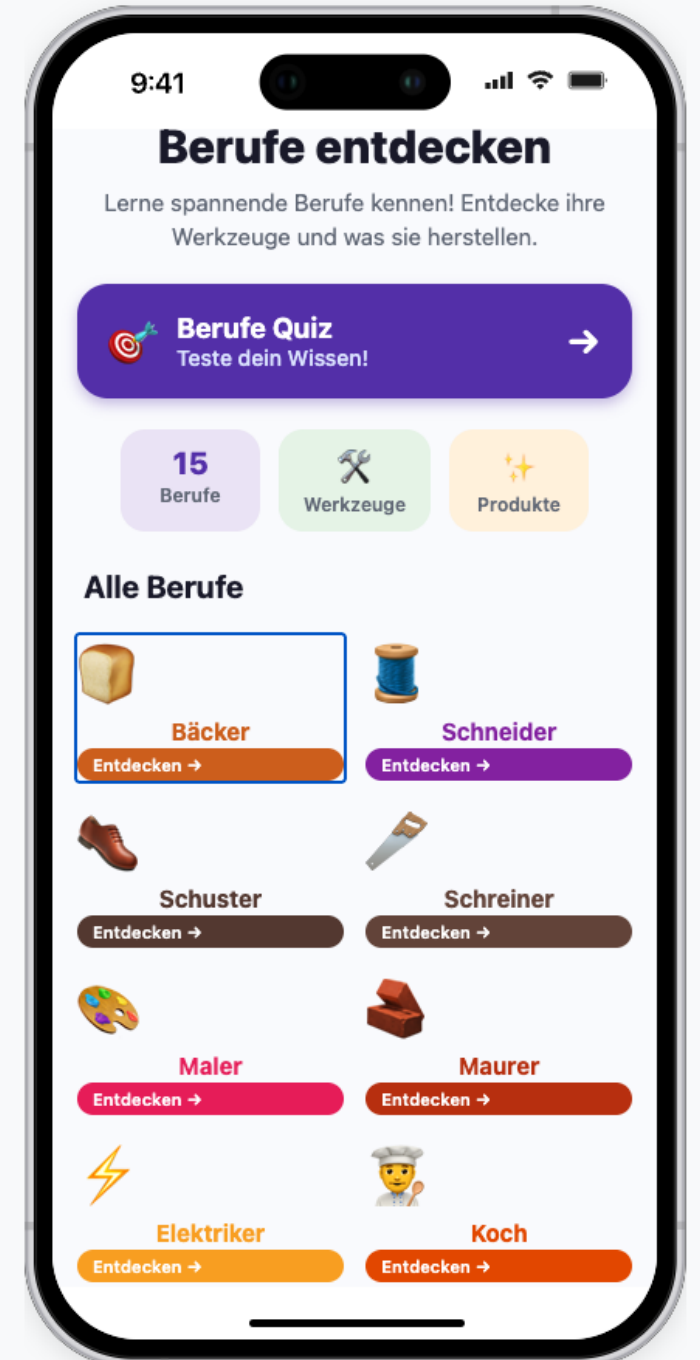
User: primary-school student preparing for an NMG test.

Problem: fragmented learning material, limited parent time, low motivation for paper-based exercises.

Idea: mobile quiz and study helper.

How this could be extended:

- Not only for a single NMG test, but for **all NMG topics**
- Share with other parents
- Enable other parents to create content and share the workload
- Build a community around it



Prompt Template

You are a helpful assistant for [target user].

Your task is to [specific task].

Context:

- The user is using this app for [situation].
- The goal is [desired outcome].

Instructions:

- Be accurate and concise.
- Ask for missing information only if essential.
- Do not invent facts.
- Keep the output practical and easy to use.

Output format:

1. [section one]
2. [section two]
3. [section three]

User input:

{{user_input}}

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Exercise

Bloom Challenge: Build a Bloom app that solves one real problem.

Form 4-5 teams.

Deliverables

Each team must present:

- app name
- target user
- one-sentence problem statement
- one live demo
- one thing they improved after testing

Submit Bloom app with link and details in Canva:

<https://canva.link/98u2fumm04vkfel>

Presentation format

- 60-second pitch
- 60-seconds demo
- 30-seconds reflection